

Tomcat 8.5 HTTPS with a JKS Keystore

Complete Windows example for the LawyersIns application - certificate creation, Tomcat configuration, validation, export, and troubleshooting.

Application	LawyersIns
Tomcat	C:/Tomcat8.5
HTTPS port	8443
Keystore	C:/Tomcat8.5/conf/lawyersins.jks
Alias	server

Important: This procedure creates a self-signed certificate. It encrypts traffic but is not automatically trusted by browsers. Use a CA-issued certificate for production.

1. Verify Java and keytool

Open PowerShell and confirm that Java and keytool are available:

```
java -version
keytool -help
```

If keytool is not on PATH, invoke it through JAVA_HOME:

```
& "$env:JAVA_HOME\bin\keytool.exe" -help
```

2. Generate the localhost JKS

Run this command for local development. The SAN extension is required by modern browsers when validating a hostname.

```
& "$env:JAVA_HOME\bin\keytool.exe" -genkeypair `
  -alias server `
  -keyalg RSA `
  -keysize 2048 `
  -sigalg SHA256withRSA `
  -validity 365 `
  -keystore "C:\Tomcat8.5\conf\lawyersins.jks" `
  -storetype JKS `
  -dname "CN=localhost, OU=Development, O=Protecture, L=Chicago, `
  ST=Illinois, C=US" `
  -ext "SAN=dns:localhost,ip:127.0.0.1"
```

Keytool prompts for the following values:

```
Enter keystore password: [enter a strong password]
Re-enter new password: [enter the same password]
Enter key password for <server>: [press Enter to reuse the password]
```

Use the same password for the private key and keystore unless you plan to configure a separate `certificateKeyPassword` in Tomcat.

3. Generate a certificate for a real hostname

Replace the example hostname and organization fields with your actual values. Every hostname used by clients must appear in SAN.

```
& "$env:JAVA_HOME\bin\keytool.exe" -genkeypair `
  -alias server `
  -keyalg RSA `
  -keysize 2048 `
  -sigalg SHA256withRSA `
  -validity 365 `
  -keystore "C:\Tomcat8.5\conf\lawyersins.jks" `
  -storetype JKS `
  -dname "CN=lawyers.example.com, OU=IT, O=Protecture, L=Chicago, `
  ST=Illinois, C=US" `
  -ext "SAN=dns:lawyers.example.com,dns:www.lawyers.example.com"
```

4. Verify the keystore

```
& "$env:JAVA_HOME\bin\keytool.exe" -list -v `
  -alias server `
  -keystore "C:\Tomcat8.5\conf\lawyersins.jks"
```

Confirm these values in the output:

- Alias name is server.
- Entry type is PrivateKeyEntry.
- Signature algorithm is SHA256withRSA.
- SubjectAlternativeName contains every required DNS name or IP address.

5. Back up Tomcat configuration

```
Copy-Item "C:\Tomcat8.5\conf\server.xml" `
  "C:\Tomcat8.5\conf\server.xml.backup"
```

6. Configure the Tomcat HTTPS connector

Open C:/Tomcat8.5/conf/server.xml. Add this Connector inside <Service name="Catalina">. If port 8443 already has an HTTPS Connector, update it instead of adding another one.

```
<Connector
  port="8443"
  protocol="org.apache.coyote.http11.Http11NioProtocol"
  maxThreads="150"
  SSLEnabled="true"
  scheme="https"
  secure="true">

  <SSLHostConfig protocols="TLSv1.2">
    <Certificate
      certificateKeystoreFile=
        "C:/Tomcat8.5/conf/lawyersins.jks"
      certificateKeystorePassword="YOUR_KEYSTORE_PASSWORD"
      certificateKeystoreType="JKS"
      certificateKeyAlias="server"
      type="RSA" />
    </SSLHostConfig>
  </Connector>
```

Replace YOUR_KEYSTORE_PASSWORD with the password used when the JKS was created. Do not leave the placeholder in server.xml.

7. Restart Tomcat

Stop and start Tomcat from PowerShell:

```
& "C:\Tomcat8.5\bin\shutdown.bat"
& "C:\Tomcat8.5\bin\startup.bat"
```

If Tomcat is managed by Eclipse, stop and restart it from the Servers view instead. Review the Tomcat console for certificate, password, path, or port-binding errors.

8. Test LawyersIns over HTTPS

Open the application in a browser:

```
https://localhost:8443/LawyersIns/
```

Or verify the connection from PowerShell:

```
curl.exe -vk "https://localhost:8443/LawyersIns/"
```

A browser warning is expected for a self-signed certificate. The warning should not occur after installing a properly chained CA-issued production certificate.

9. Export the public certificate

```
& "$env:JAVA_HOME\bin\keytool.exe" -exportcert `
  -alias server `
  -keystore "C:\Tomcat8.5\conf\lawyersins.jks" `
  -rfc `
  -file "C:\Tomcat8.5\conf\lawyersins.cer"
```

The exported file is `C:/Tomcat8.5/conf/lawyersins.cer`. Only distribute the public `.cer` file. Never distribute the JKS because it contains the private key.

10. Production certificate request

Generate a CSR for submission to a trusted certificate authority:

```
& "$env:JAVA_HOME\bin\keytool.exe" -certreq `
  -alias server `
  -keystore "C:\Tomcat8.5\conf\lawyersins.jks" `
  -file "C:\Tomcat8.5\conf\lawyersins.csr" `
  -ext "SAN=dns:lawyers.example.com,dns:www.lawyers.example.com"
```

After the CA returns the certificate and chain, import the CA root/intermediate certificates first, then import the server certificate using the same server alias.

Troubleshooting

Message	Resolution
Keystore was tampered with, or password was incorrect	Confirm certificateKeystorePassword and the JKS password. Confirm that the key password is the same, or configure certificateKeyPassword.
Alias does not identify a key entry	Run keytool -list -v and verify that alias server is a PrivateKeyEntry.
FileNotFoundException for lawyersins.jks	Use the absolute path C:/Tomcat8.5/conf/lawyersins.jks and verify Tomcat service account read permission.
Address already in use: bind	Another process is using port 8443. Stop that process or select another unused port.
Browser reports hostname mismatch	Regenerate or replace the certificate so SAN includes the exact hostname used in the browser URL.
HTTPS works but LawyersIns does not open	Verify the application context is /LawyersIns and inspect Tomcat startup logs for deployment errors.

Security checklist

- Use a unique strong keystore password and restrict access to server.xml.
- Never commit the JKS, its password, or the private key to source control.
- Use a CA-issued certificate for production and monitor its expiration date.
- Permit only required TLS versions and keep Java and Tomcat patched.
- Back up the JKS securely before replacing or renewing certificates.

Reference

Apache Tomcat 8.5 SSL/TLS Configuration How-To:

<https://tomcat.apache.org/tomcat-8.5-doc/ssl-howto.html>